MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12B4A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, miniature, low-Mu

Outline	---	6-3 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	k	g	hct	h	h	nc	g	nc	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	eb	Eb	Ec	(A) Ehk	Rg	Pp	(A) Ik	ik	(A) Alt
Unit:	V	v	Vdc	Vdc	v	Meg	W	mAdc	ma	ft
Maximum:	6.9 or 13.9	1,000 (note 2)	550	0	(Note 8)	0.47 (note 3)	6 (note 4)	30 (notes 2&4)	105 (note 2)	(Note 1)
Minimum:	5.7 or 11.4	---	---	-250 (note 2)	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	150	-17.5	---	---	---	---	---	---

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Rg = 0.5 Meg (see note 5)	0.65	II	Ic	0	-1.0	µAdc
1306	Transconductance (1)		0.65	II	Sm	5,000	7,600	µmhos
1256	Electrode current (1) (anode)	Ec = -38 Vdc; Rp = 0.1 Meg	0.65	II	Ib	---	200	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		2.5	I	If	570	630	mA
1336	Heater-cathode leakage		2.5	I	Ihk	---	20	µAdc
1256	Electrode current (2) (anode)		2.5	I	Ib	20	48	mAdc
(A) 2201	Noise and microphonics	Ef = 12.6 Vdc; Ebb = 300 Vdc; Ecal = 75 mVrms; Ec = 0; Rg = 1.0 Meg; Rl = 0.05 Meg; Rk = 1,000 ohms (see note 7)	2.5	I	EB	---	17	vu
1306	Transconductance (2)	Ef = 5.7 V	2.5	S3	Sm	4,500	---	µmhos
---	High-voltage arcover	Ef = 0 (see note 6)	2.5	S3	ep	1,000	---	v
1316	Amplification factor		---	---	Mu	5.8	7.2	---
(A) 1121	Base strain		---	---	---	---	---	---
(A) 2126	Glass strain		4.0	I	---	---	---	---
(A) 1211	Insulation of electrodes		4.0	S3	---	---	---	---
(A) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Ef = 12.6 V; Eb = 180 Vdc; Ec = 0; Ehk = -180 Vdc; Rg = 2.2 Meg; Rk = 550 ohms (see note 7)	---	---	---	---	---	---
---	Intermittent life end points (500 hours)	Transconductance (1)	---	---	Sm	4,100	---	µmhos
		Total grid current	---	---	Ic	---	-3.0	µAdc

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. (A) The duty cycle of the voltage pulse shall not exceed 15 percent of one cycle.
3. This value is for class A amplifier operation with fixed bias. With cathode bias, Rg may be 2.2 Meg (max).
4. In stages operating with grid-leak bias, an adequate cathode-bias resistor or other suitable means is required to protect the tube in the absence of excitation.
5. This test to be performed at the conclusion of the holding period.
6. Tie grid, the no connection pins, and one side of the heater to positive terminal of cathode resistor. Apply high-voltage pulse to anode and tie cathode to ground through 100 ohms. The tube under test shall be vibrated continuously or tapped, at a peak acceleration level of approximately 2.5G, during the high-voltage arc test. The pulse generator shall be capable of supplying a high voltage pulse of approximately 10 microseconds duration at a frequency of 15.75 kHz. The anode voltage shall be increased slowly until the specified minimum anode voltage limit is reached. Two or more arc indications at an anode voltage level below the specified minimum limit shall be cause for rejection. In event one arc indication is obtained, the tube under test shall be operated for 5 seconds (minimum), at the voltage level at which the arc occurred, prior to an anode voltage increase. An arc shall be defined as a peak forward surge current equal to or greater than a peak current of 20 ma. The arc indicator shall consist of a suitable indicating device connected across a 100-ohm indicating resistor in the cathode circuit.
7. *Apply Ef between pins 4 and 5* (series heater connection).
8. Heater positive with respect to cathode: dc component = 100 volts; total dc and peak = 200 volts. Heater negative with respect to cathode: total dc and peak = 200 volts.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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